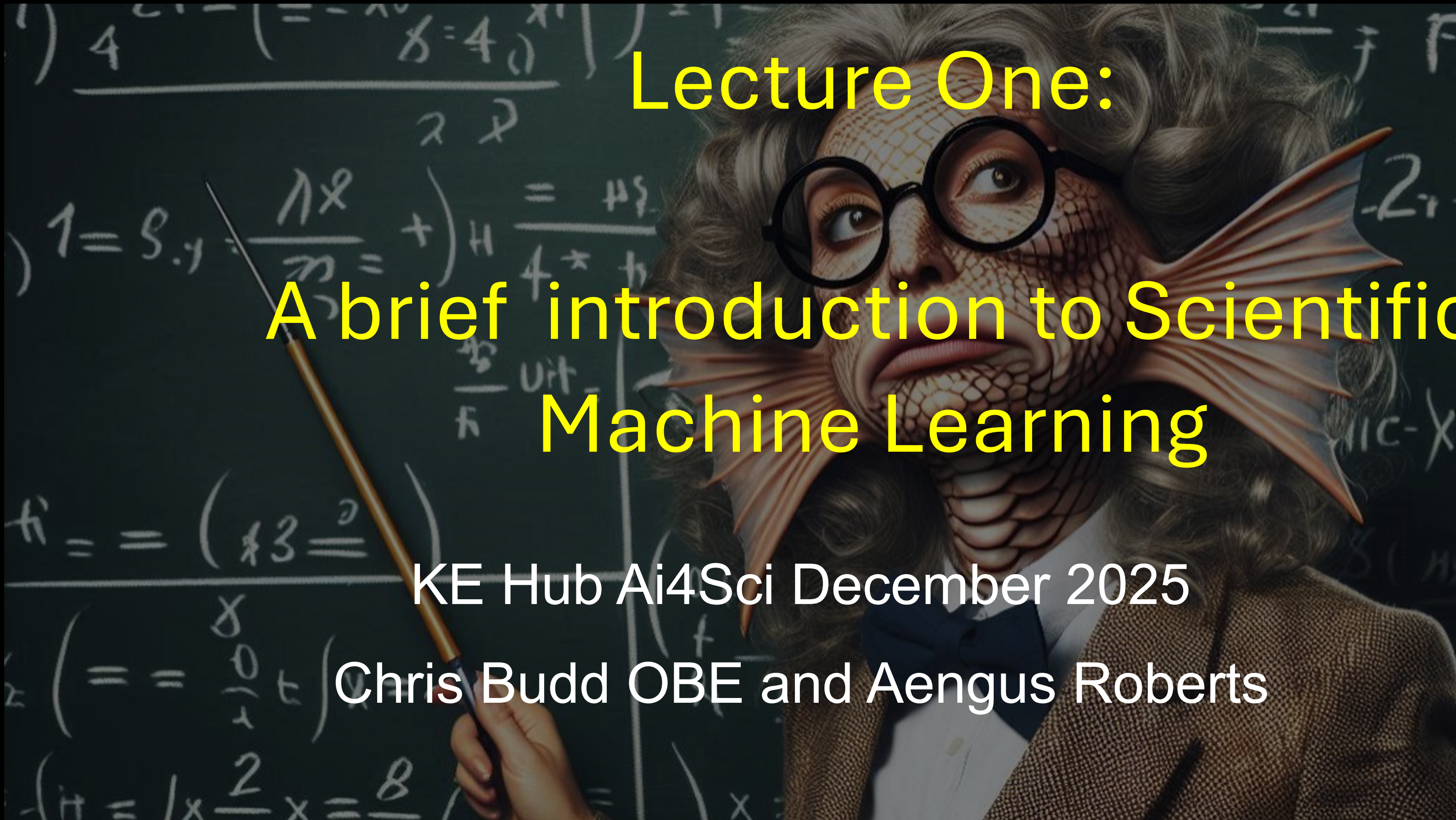


Lecture One:

A brief introduction to Scientific Machine Learning

KE Hub Ai4Sci December 2025

Chris Budd OBE and Aengus Roberts

m4DL

UNIVERSITY OF BATH

www.bbc.co.uk/news/science-environment-67383755

NEWS

Home | InDepth | Israel-Gaza war | War in Ukraine | Climate | UK | World | Business | Politics | Culture | More

Science & Environment

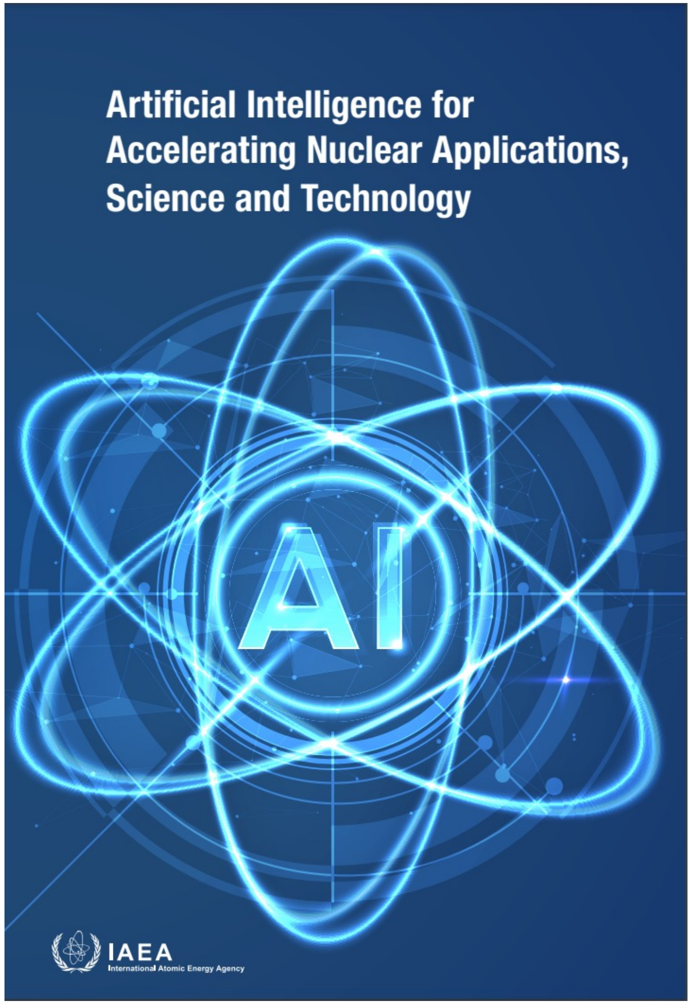
AI could predict hurricane landfall sooner - report

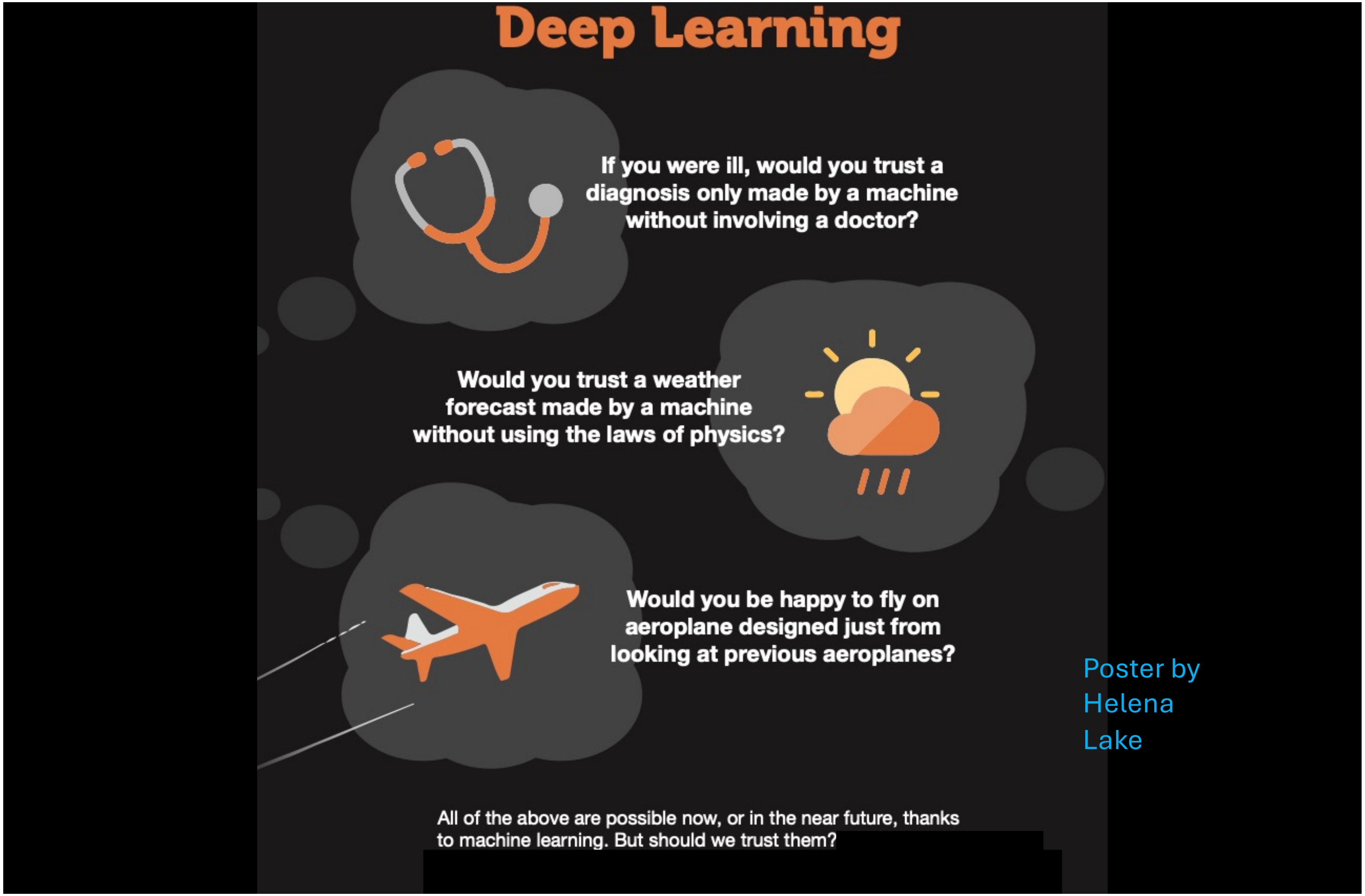
14 November 2023 · 45 Comments

AI In the News

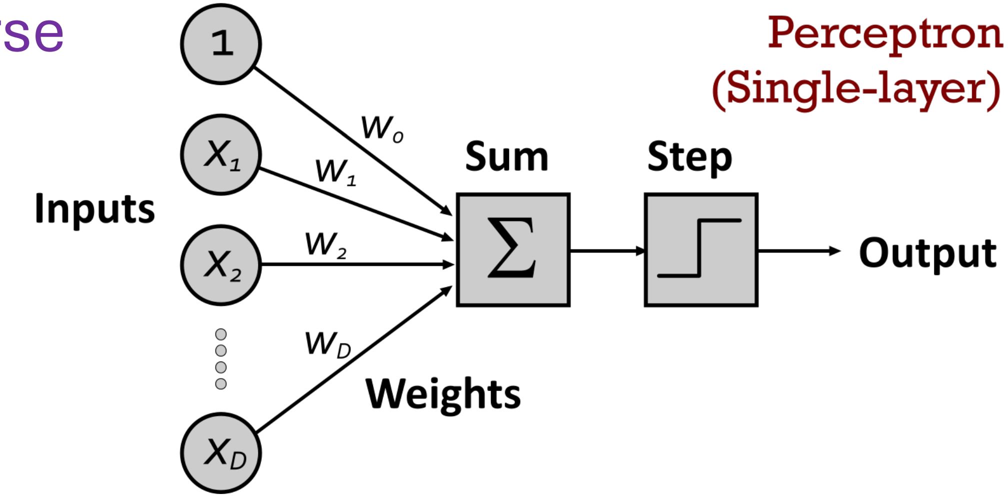
Makes ‘better prediction of Hurricane Lee’ than any other method

How good is the hype?


Artificial Intelligence for
Accelerating Nuclear Applications,
Science and Technology
AI
IAEA
International Atomic Energy Agency


Deep Learning
If you were ill, would you trust a diagnosis only made by a machine without involving a doctor?
Would you trust a weather forecast made by a machine without using the laws of physics?
Would you be happy to fly on aeroplane designed just from looking at previous aeroplanes?
All of the above are possible now, or in the near future, thanks to machine learning. But should we trust them? [redacted]
Poster by Helena Lake

Structure of the mini-course



1. Introduction

2. Architectures

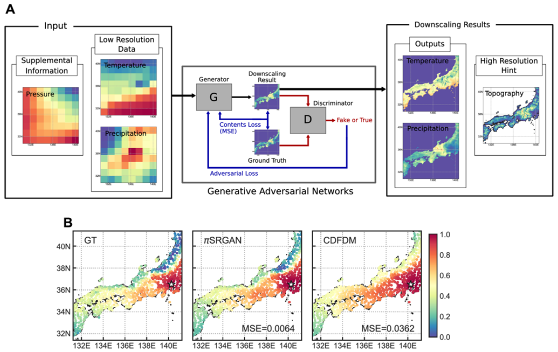
3. ML for differential equations

4. Neural operators and the weather

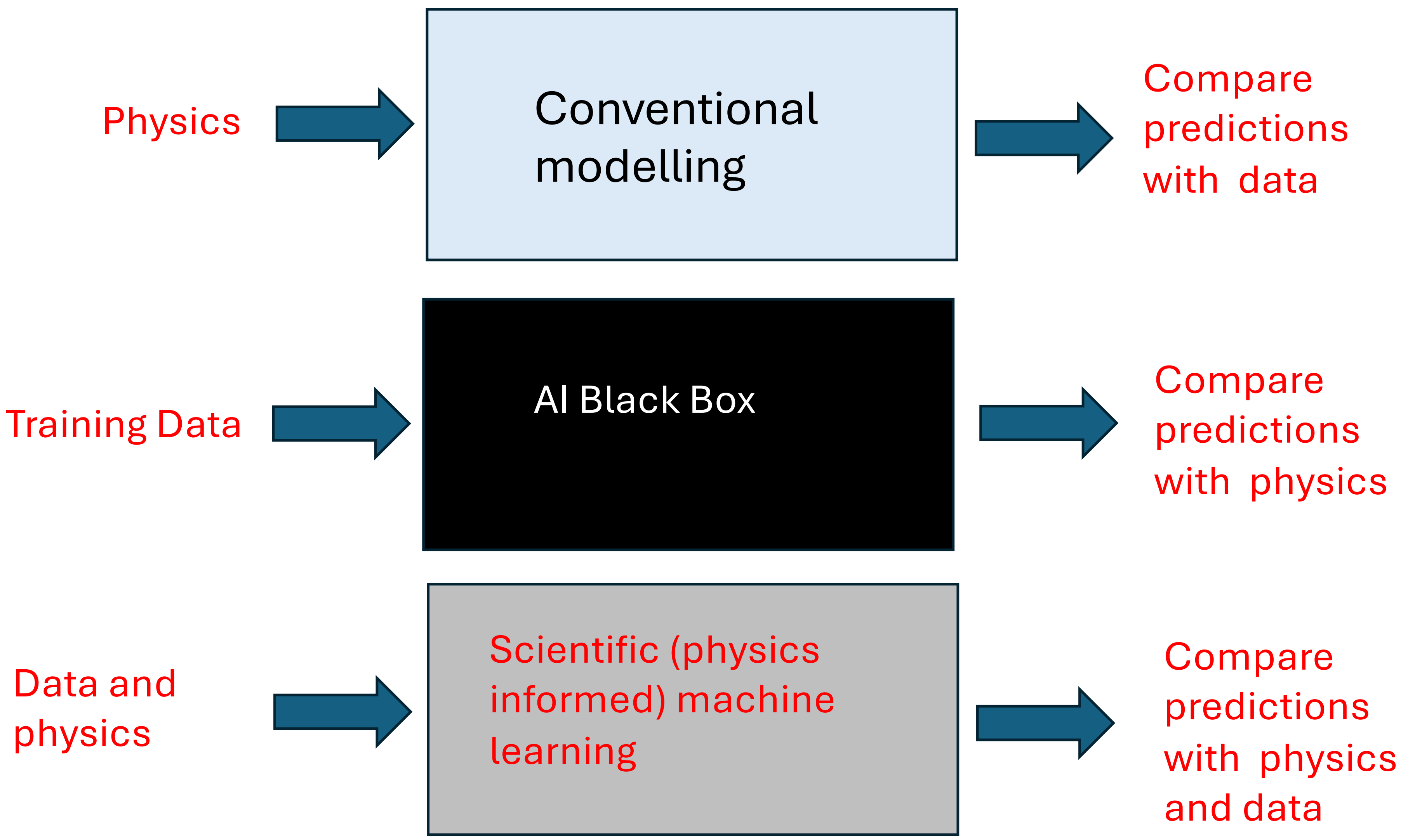
5. Neural ODES

6 Case Study: PINNS

We are seeing a major impact of machine learning mathematics, physics, engineering, and business



- Classification and identification
- Solution of differential equations (ODEs and PDEs)
- Simulation of differential operators
- Solution and regularization of inverse problems
- Prediction and time series analysis
- Classification and generation of images (down scaling)
- CAD
- Forecasting the weather (from data alone!)

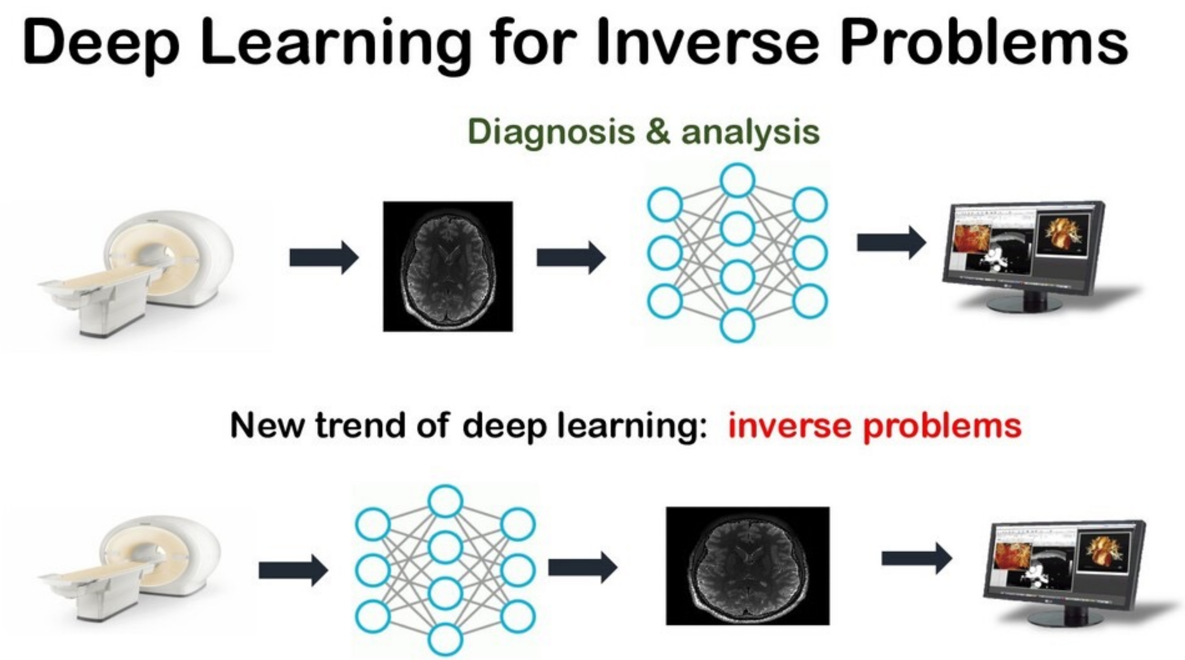

Physics
Conventional modelling
Compare predictions with data
Training Data
AI Black Box
Compare predictions with physics
Data and physics
Scientific (physics informed) machine learning
Compare predictions with physics and data

Scientific Machine Learning

Aims:

- Derive

- Effective
- Efficient
- Transparent
- Reliable
- Unbiased (data, inductive, learning)



Machine learning methods informed by physics and other disciplines

- Prove (where possible) rigorous results on their accuracy, stability, and convergence

- Apply them to reliably solve important scientific and engineering problems

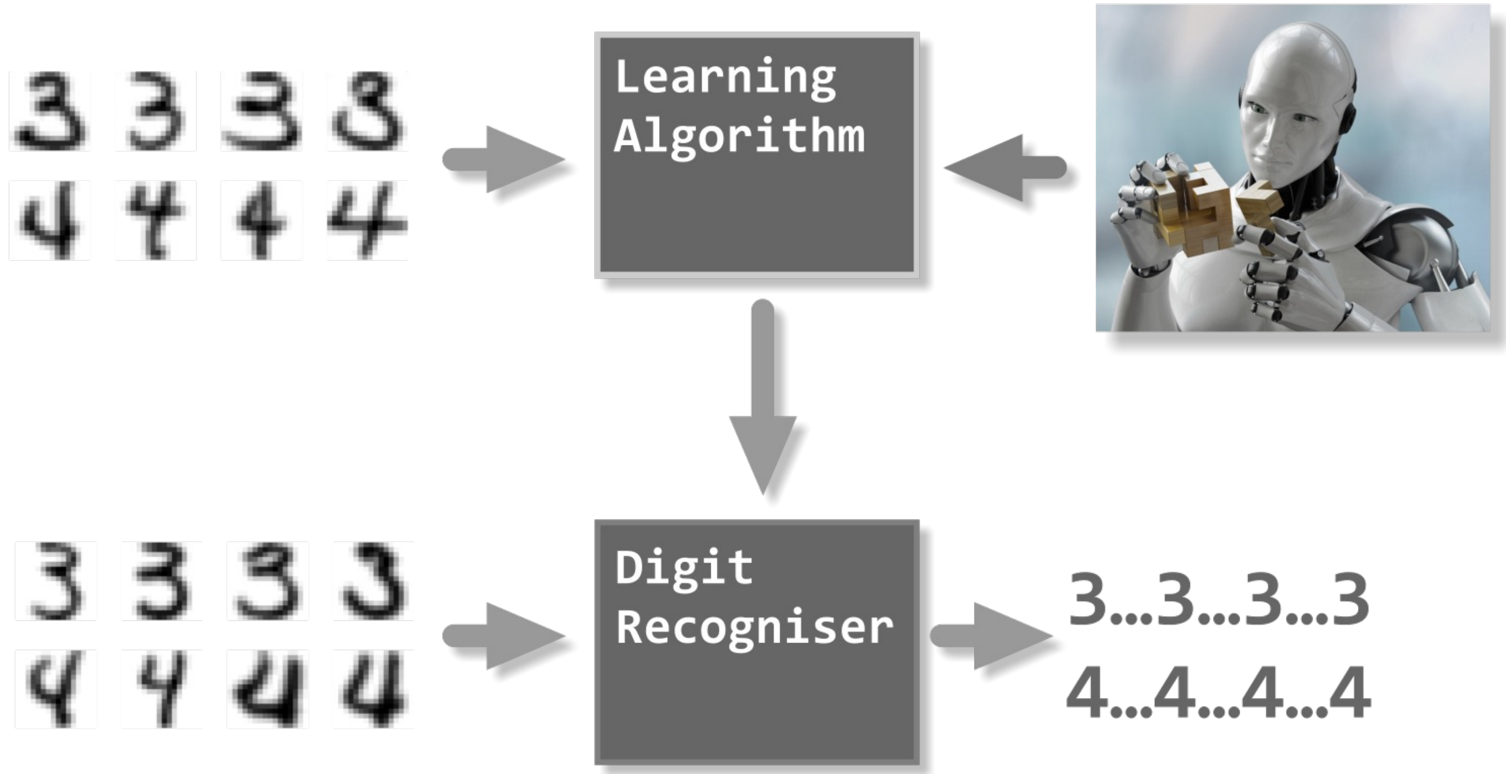
A brief introduction to the world of (scientific) machine learning

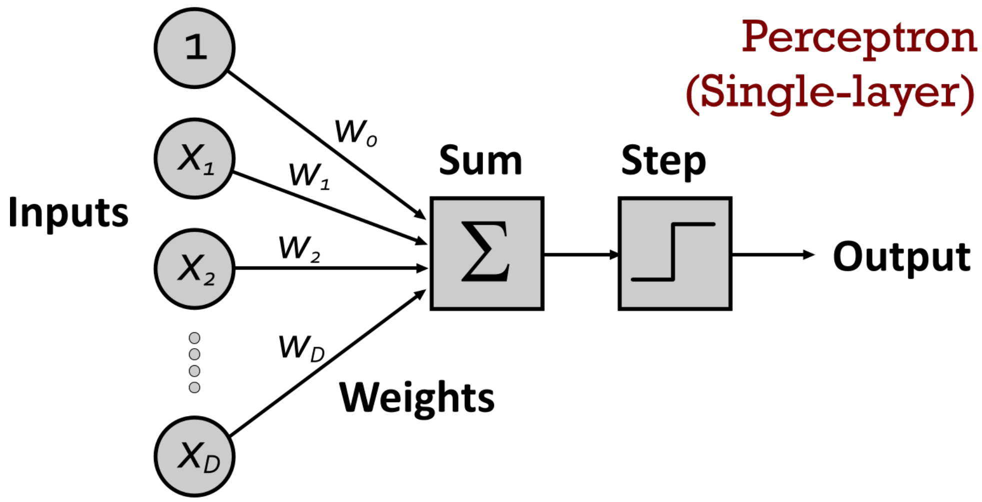
Image classification/Prediction

Probability distribution of images

Neural net trained on samples from the set of images

Image label

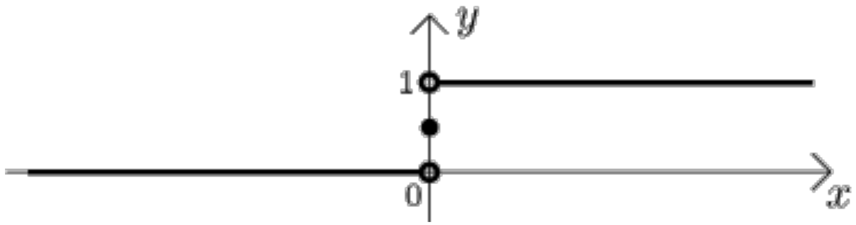

Learning Algorithm
Digit Recogniser
3...3...3...3
4...4...4...4



Breakthrough in classification came with the invention of the neural net (1963)

$$Out = H\left(\sum_{i=1}^{D} w_i \ X_i - C\right)$$

$$H(x) = \begin{cases} 1, & x > 0; \\ \frac{1}{2}, & x = 0; \\ 0, & x < 0. \end{cases}$$



Example: Can you tell a cat from a dog?

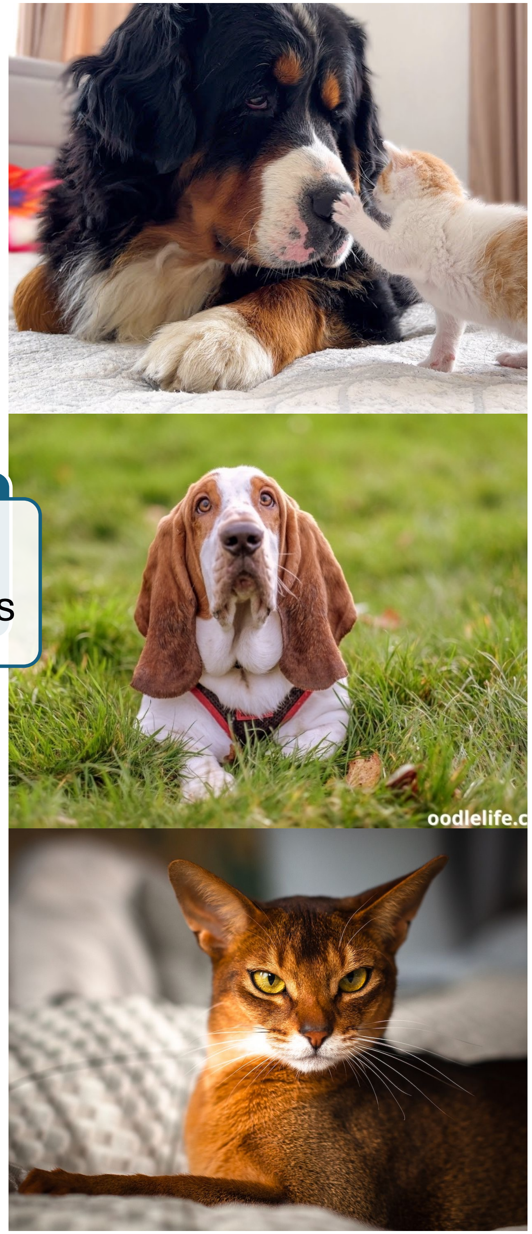
Make

two

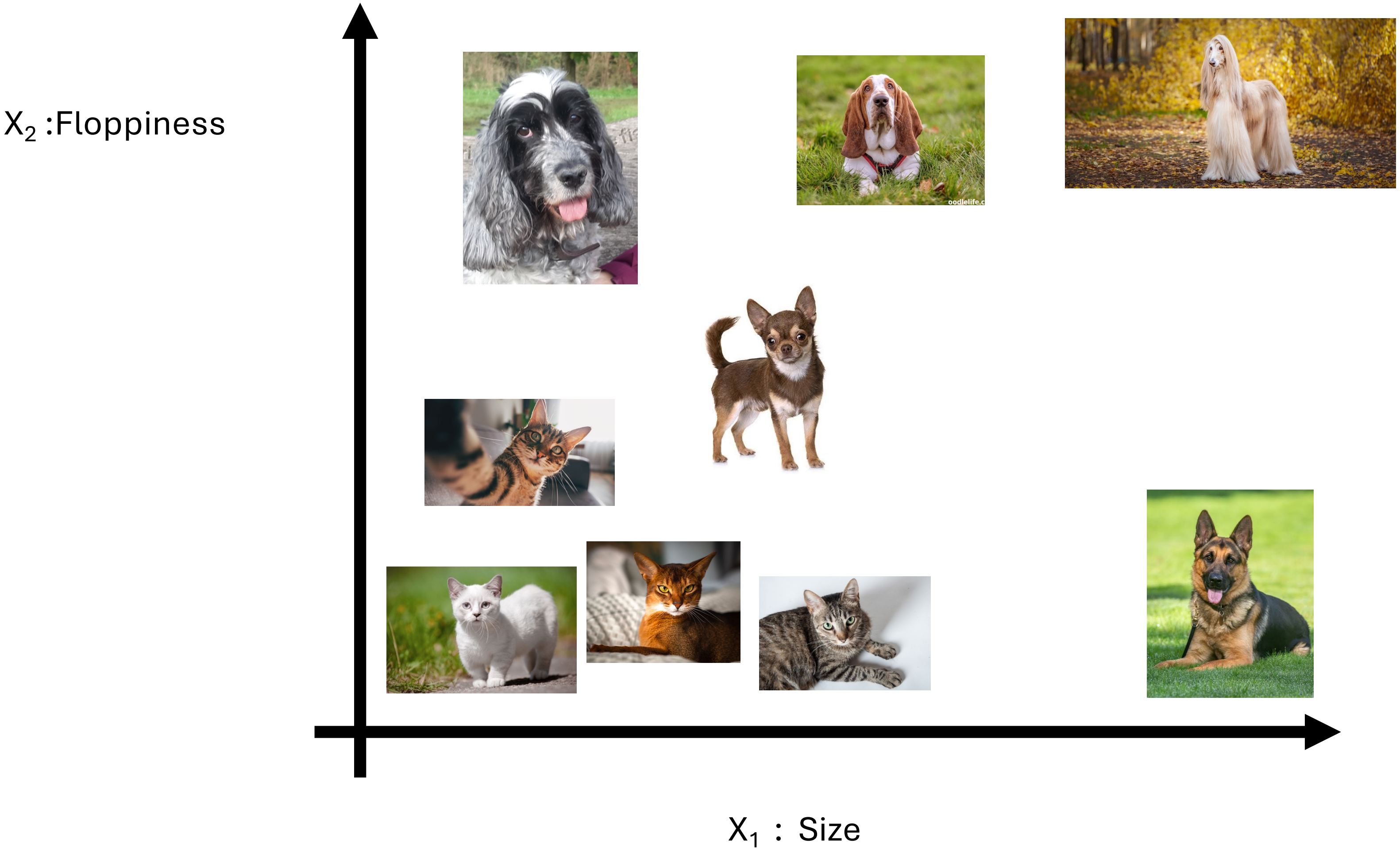
measurements

X_1 : Size

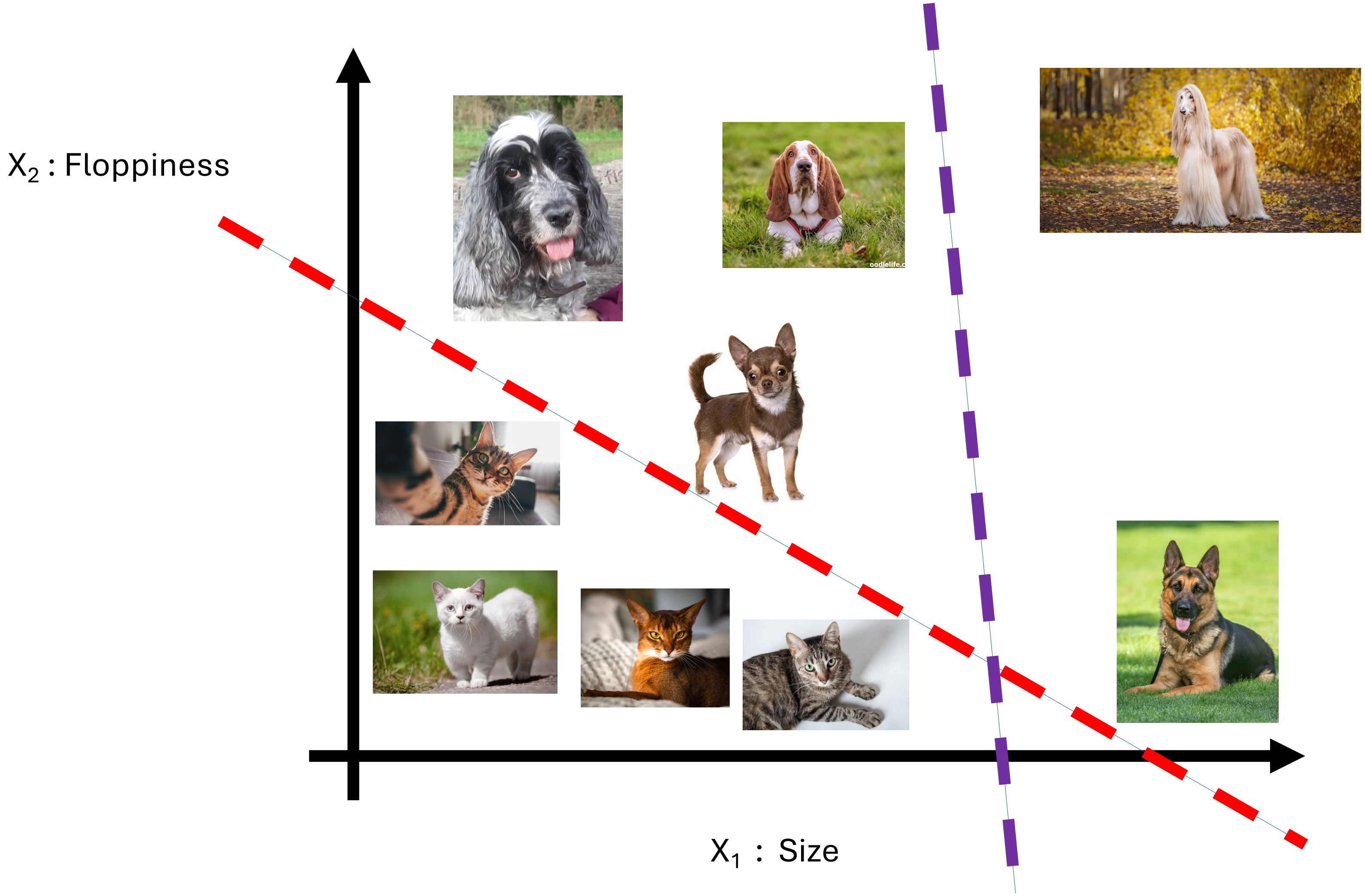
X_2 : Floppiness



Measure size and floppiness for lots of cats and dogs and plot



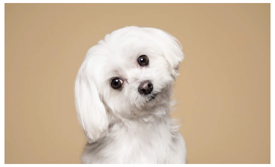
Draw a line which separates cats from dogs



What is going on?

Equation for the line is: $$w_1 X_1 + w_2 X_2 - C = 0$$

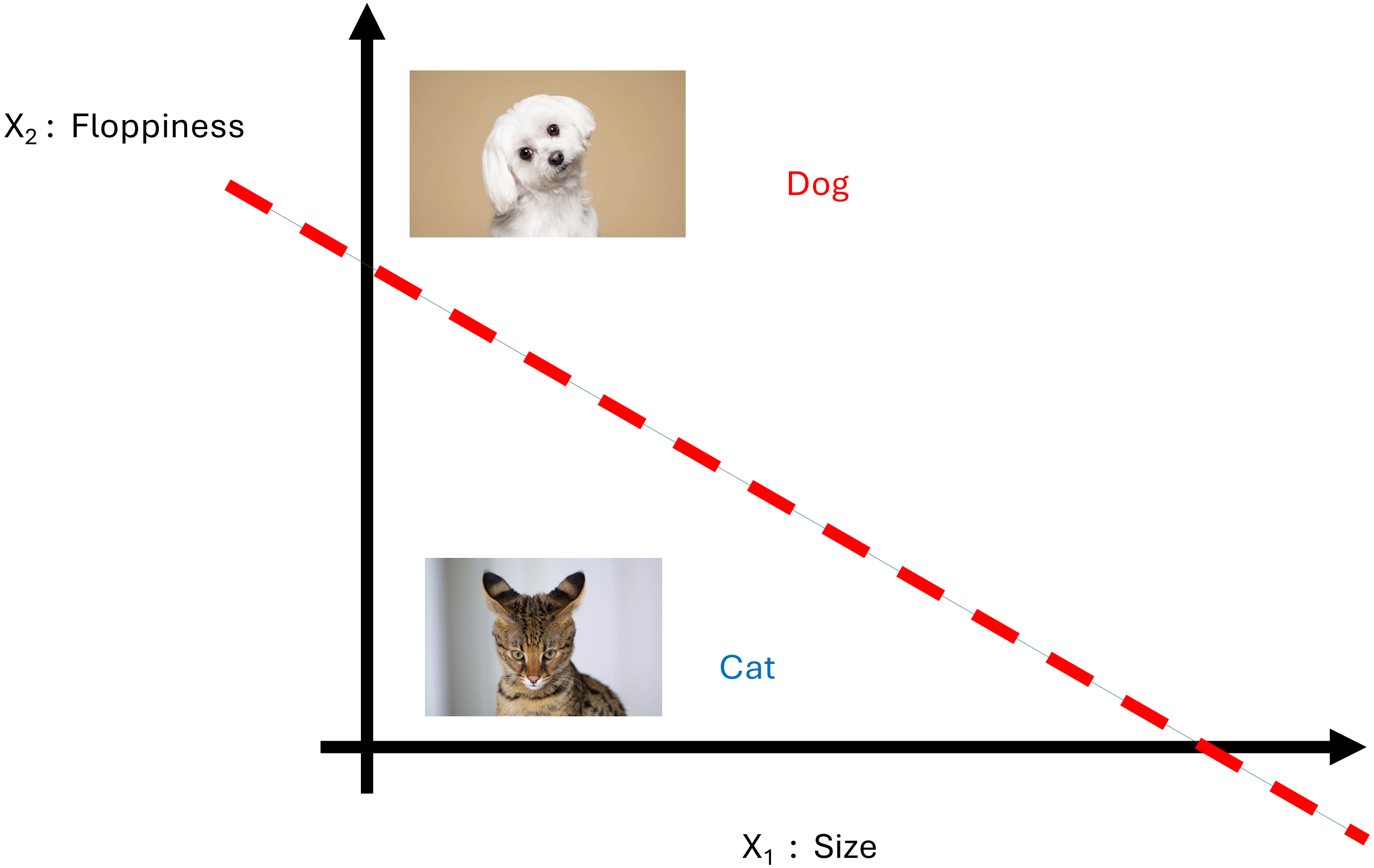
$w_1 X_1 + w_2 X_2 - C > 0$: Dog



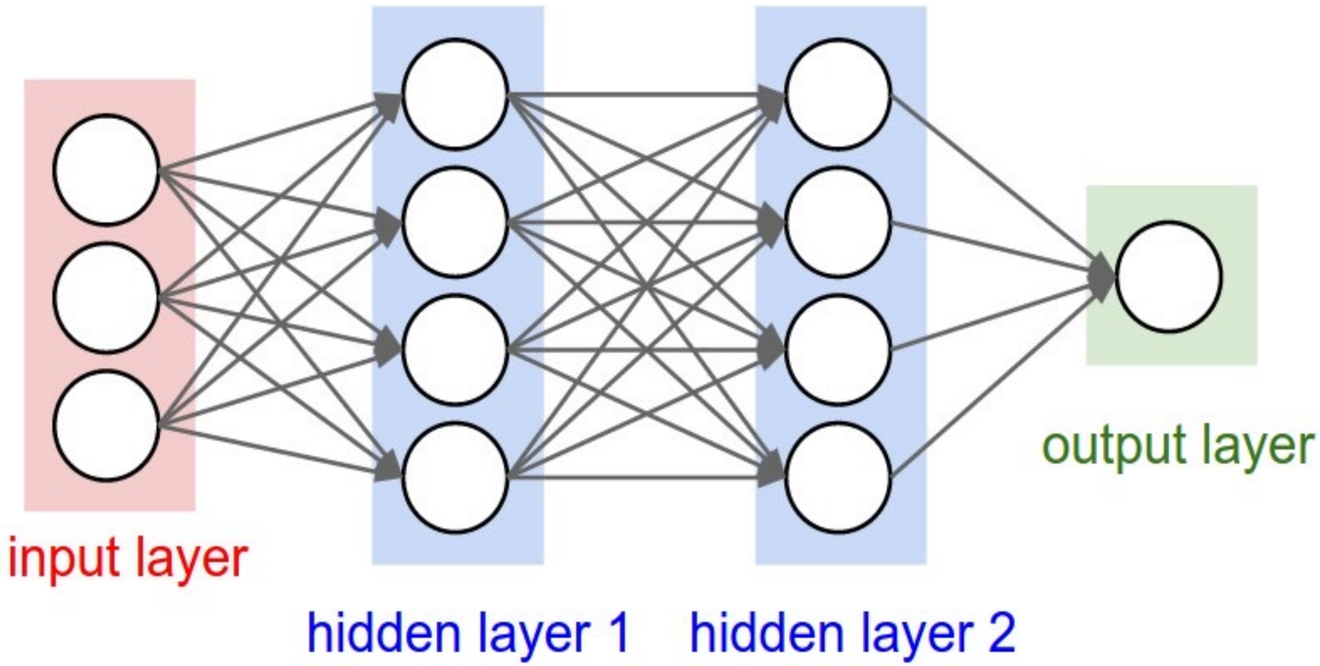
$w_1 X_1 + w_2 X_2 - C < 0.$: Cat

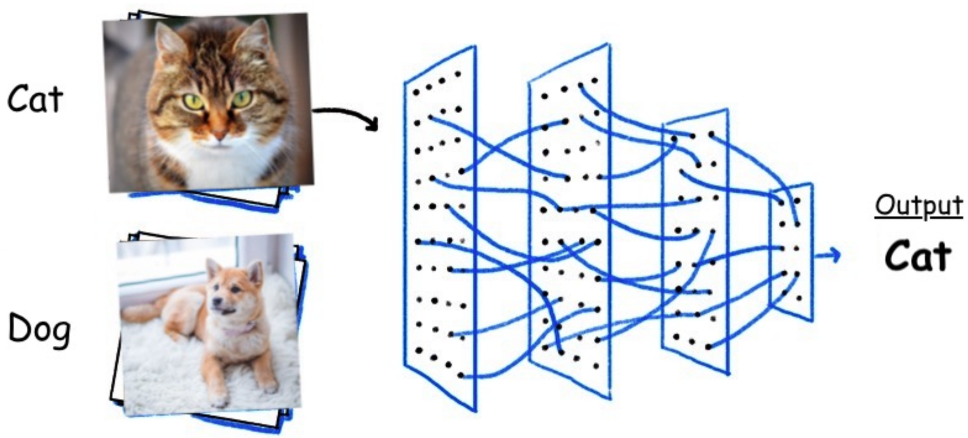
Train the weights w_1, w_2, C on the data to give the right result

Can now tell a cat from a dog

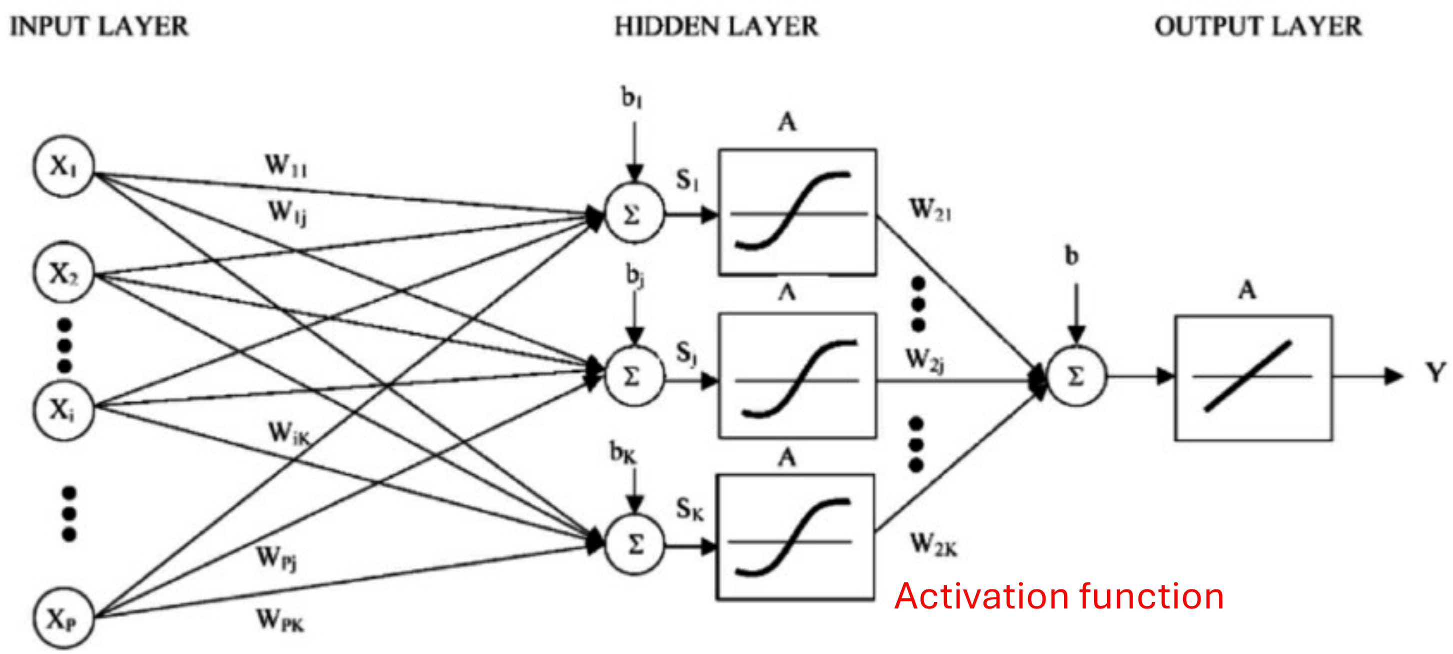


Deep neural nets (2010) use many more layers and can be used to make much more complex decisions





Cat
Dog
Output
Cat



Feed forward neural network

Or more mathematically, using a notation we use later on:

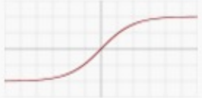
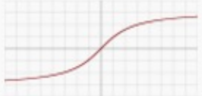
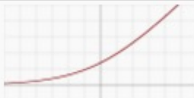
$$y_{j+1} = \sum_{i=0}^{W-1} c_i \; \sigma_{i,j} \left(a_{i,j} \; y_j + b_{i,j} \right)$$

Input: $y_0 = x$. Output: $y(x) = y_L$

W: Width L: Depth

$\sigma_{i,j}$: Activation functions. (usually nonlinear)

θ: Set of All of the parameters

Name	Plot	Equation	Derivative
Identity		$f(x)=x$	$f'(x)=1$
Binary step		$f(x)=\begin{cases}0 & \text{for } x<0\\ 1 & \text{for } x\geq 0\end{cases}$	$f'(x)=\begin{cases}0 & \text{for } x\neq 0\\ ? & \text{for } x=0\end{cases}$
Logistic (a.k.a Soft step)		$f(x)=\frac{1}{1+e^{-x}}$	$f'(x)=f(x)(1-f(x))$
TanH		$f(x)=\tanh(x)=\frac{2}{1+e^{-2x}}-1$	$f'(x)=1-f(x)^2$
ArcTan		$f(x)=\tan^{-1}(x)$	$f'(x)=\frac{1}{x^2+1}$
Rectified Linear Unit (ReLU)		$f(x)=\begin{cases}0 & \text{for } x<0\\ x & \text{for } x\geq 0\end{cases}$	$f'(x)=\begin{cases}0 & \text{for } x<0\\ 1 & \text{for } x\geq 0\end{cases}$
Parameteric Rectified Linear Unit (PReLU)[2]		$f(x)=\begin{cases}\alpha x & \text{for } x<0\\ x & \text{for } x\geq 0\end{cases}$	$f'(x)=\begin{cases}\alpha & \text{for } x<0\\ 1 & \text{for } x\geq 0\end{cases}$
Exponential Linear Unit (ELU)[3]		$f(x)=\begin{cases}\alpha(e^x-1) & \text{for } x<0\\ x & \text{for } x\geq 0\end{cases}$	$f'(x)=\begin{cases}f(x)+\alpha & \text{for } x<0\\ 1 & \text{for } x\geq 0\end{cases}$
SoftPlus		$f(x)=\log_e(1+e^x)$	$f'(x)=\frac{1}{1+e^{-x}}$

Need to find the best values for the Parameters

Achieve this by training the NN using a training set of data and a measure of success

Can take many hours of computer time on a GPU.

With a vast expense of energy!!

Mathematical Ingredients of the training process

Loss function $L(W)$: How close is the answer

Back propagation: How does the answer depend on changes in the parameters.

Achieved by using automatic differentiation

PyTorch does this with ease!

Optimisation: Choose the parameters to

minimize the loss over (a subset of the) training data

Validation: Test on separate data to avoid over fitting to the answer

Optimise the weights using optimisation methods:

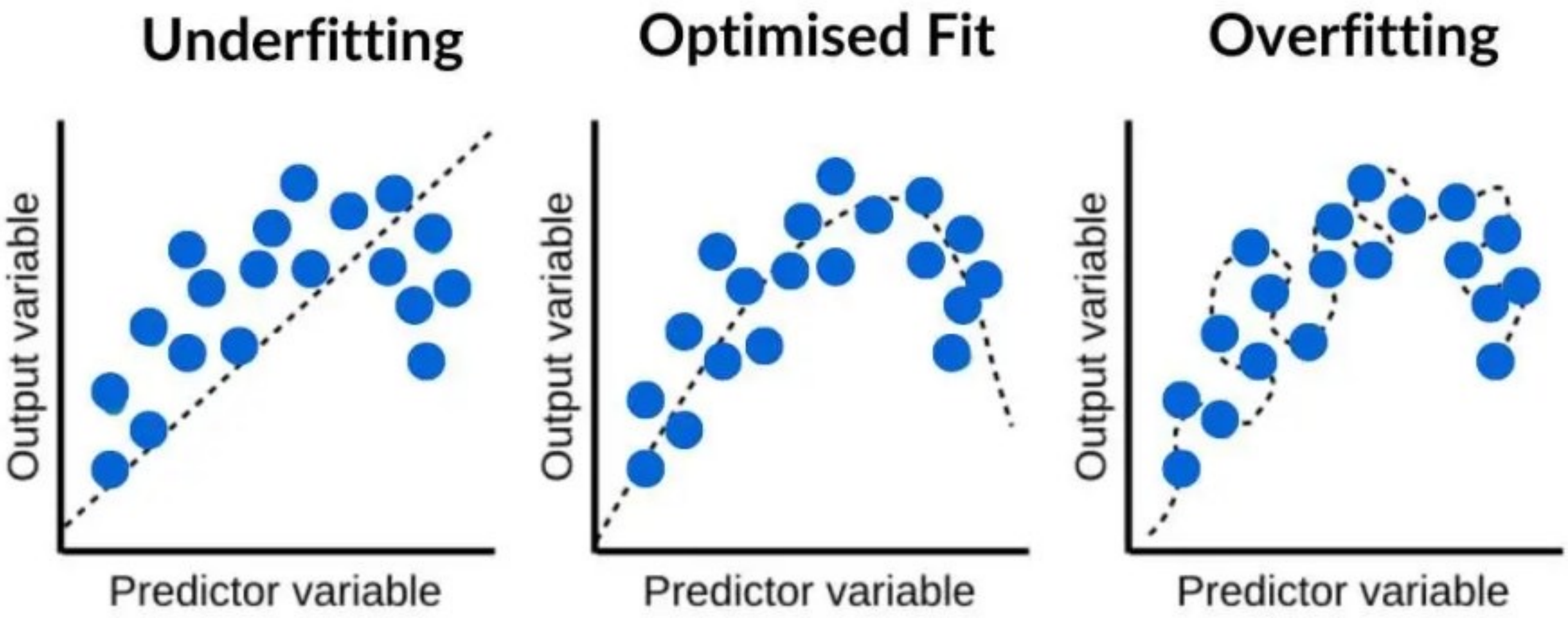
Eg. stochastic gradient descent, ADAM, quasi-Newton, simulated annealing, Monte-Carlo Tree Search (MCTS)

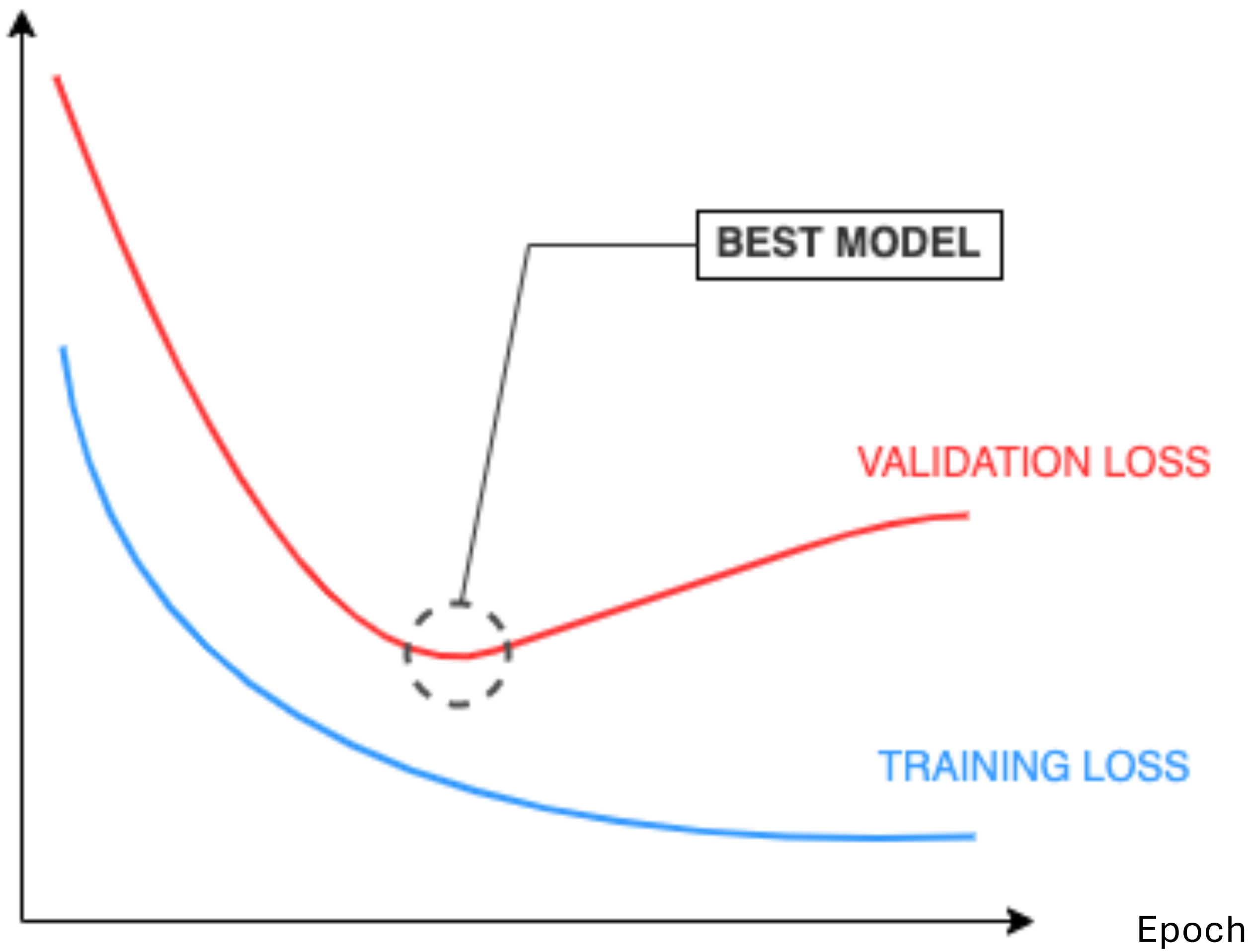
Implemented in eg. PyTorch

(see the Case studies)

Tensor Flow/Keras

AlphaZero used reinforcement learning over 700,000 games with MCTS optimisation


Underfitting
Optimised Fit
Overfitting
Output variable
Predictor variable
Output variable
Predictor variable
Output variable
Predictor variable


BEST MODEL
VALIDATION LOSS
TRAINING LOSS
Epoch

- Supervised training

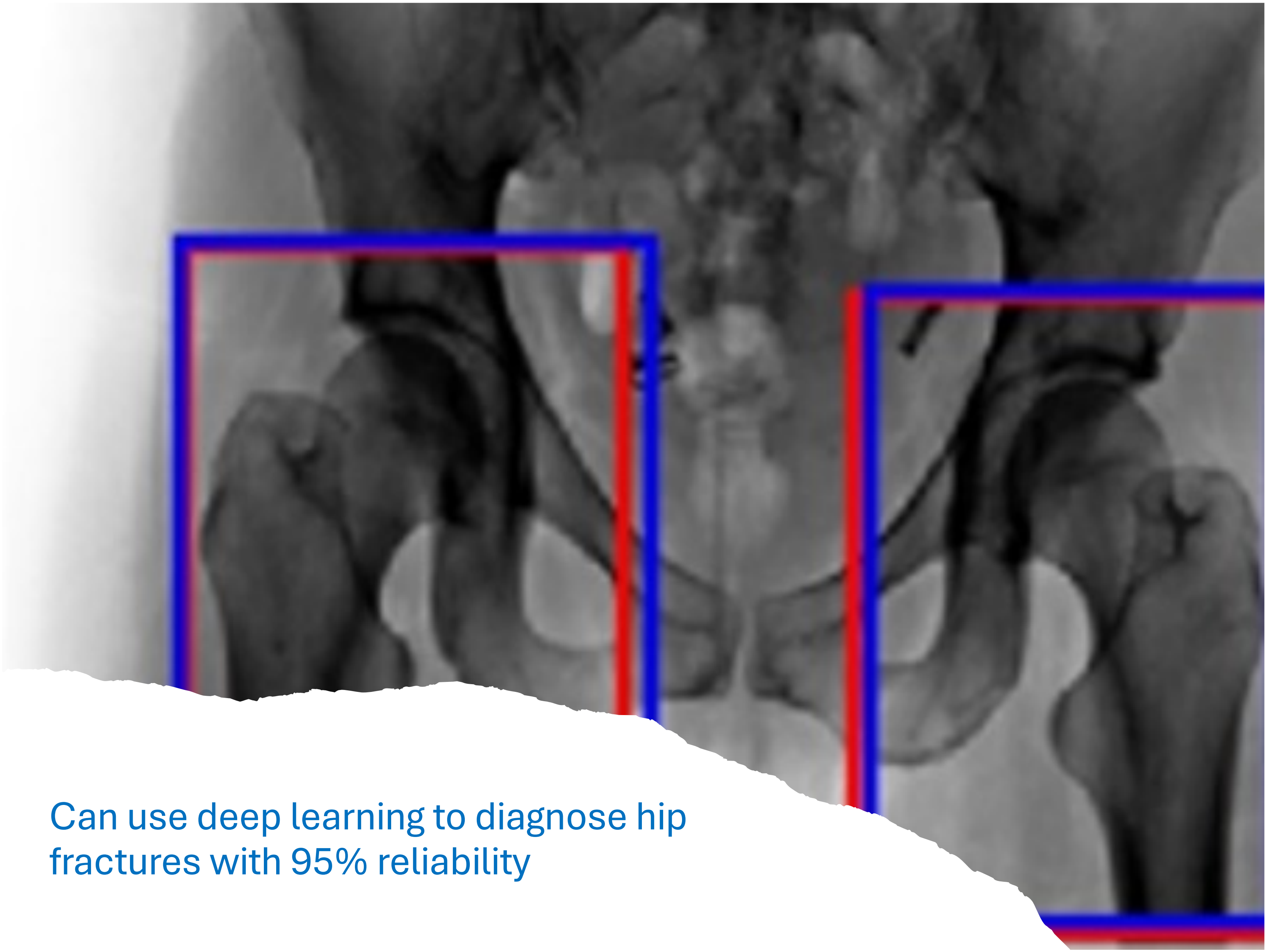
Have a labled training set

eg. cats + dogs, musical genres, Inverse problems, approximation

- Reinforcement training

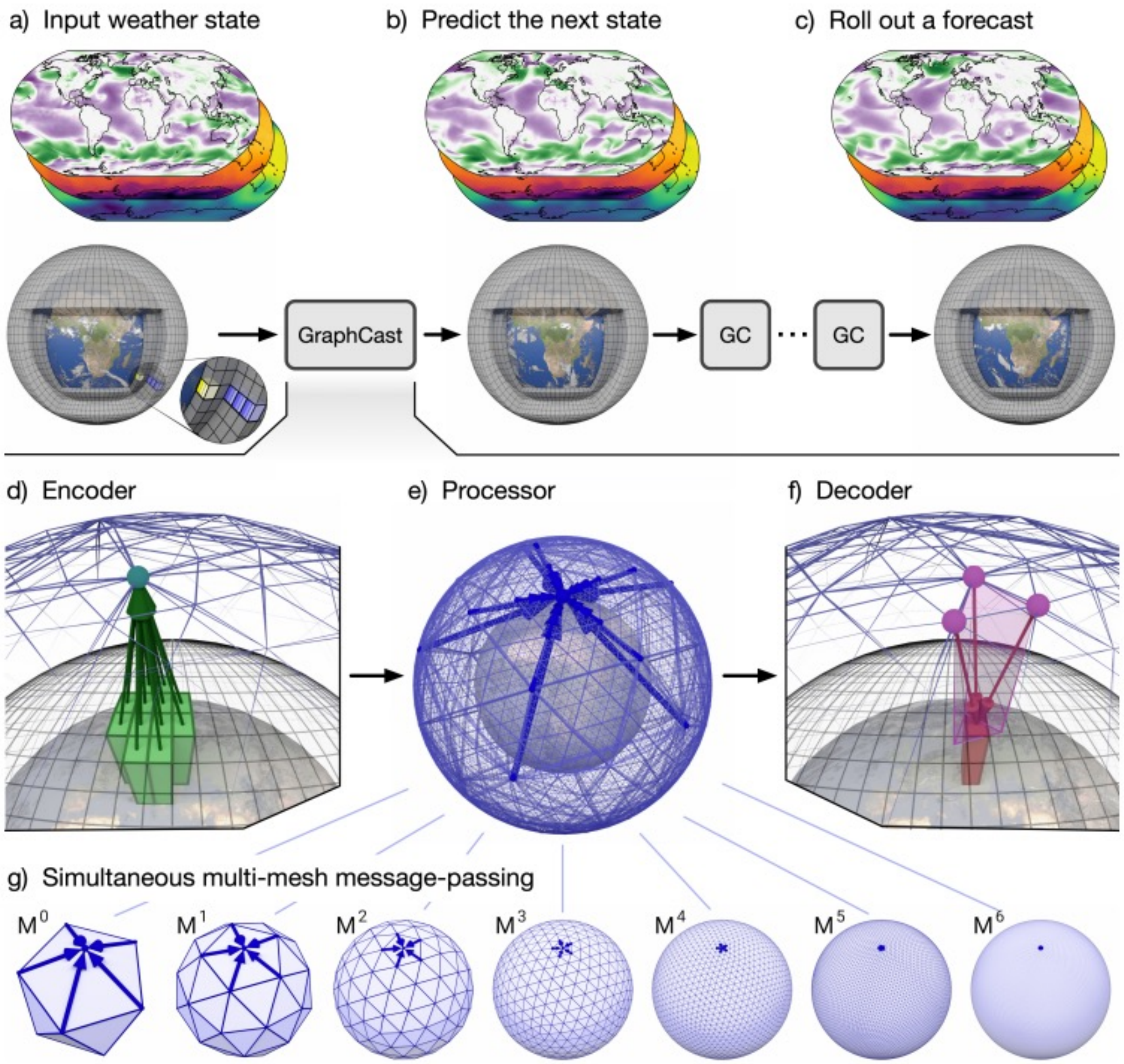
Have a measure of success

eg. winning GO, solving an equation (eg. a PINN), variational method



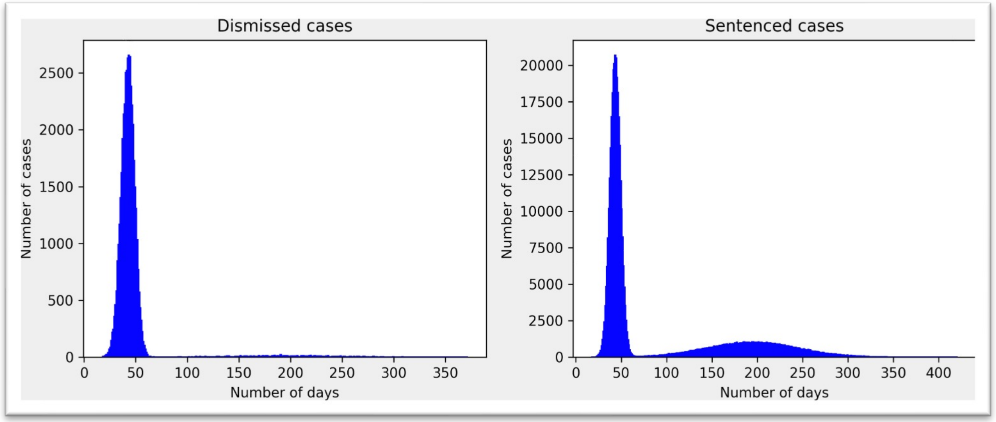
Can use deep learning to diagnose hip fractures with 95% reliability

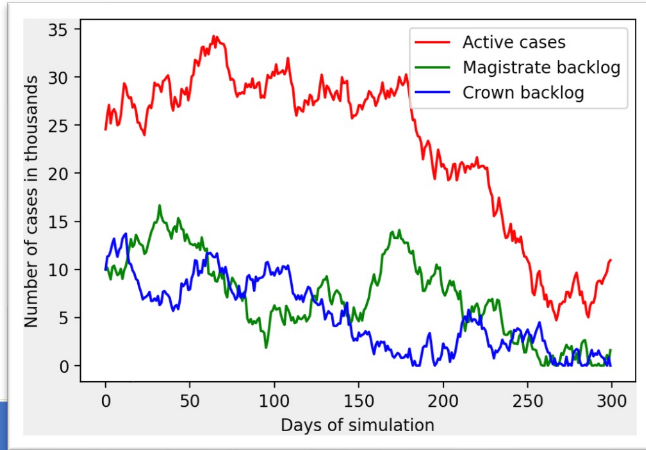
Weather forecasting trained on the ERA5 Data Set



Maths and AI for Justice

NN can predict case volume and duration in the judicial system based on case characteristics



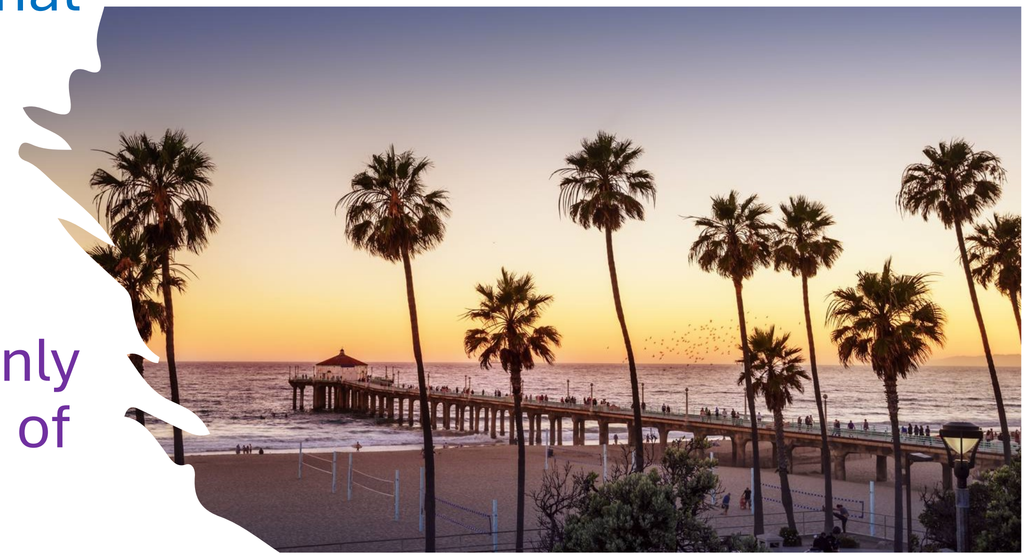


Time distribution: Lognormal
Time spent in summoned: 34 days
Summoned to Exit: 9 days
Time spent in Sentencing: 7 days

Data: the national statistics for the Criminal court statistics quarterly (April - June 2023)

But ..Can a weather forecast trained on Californian data predict a storm in the UK?

Learning only from data makes it hard to make decisions away from that data



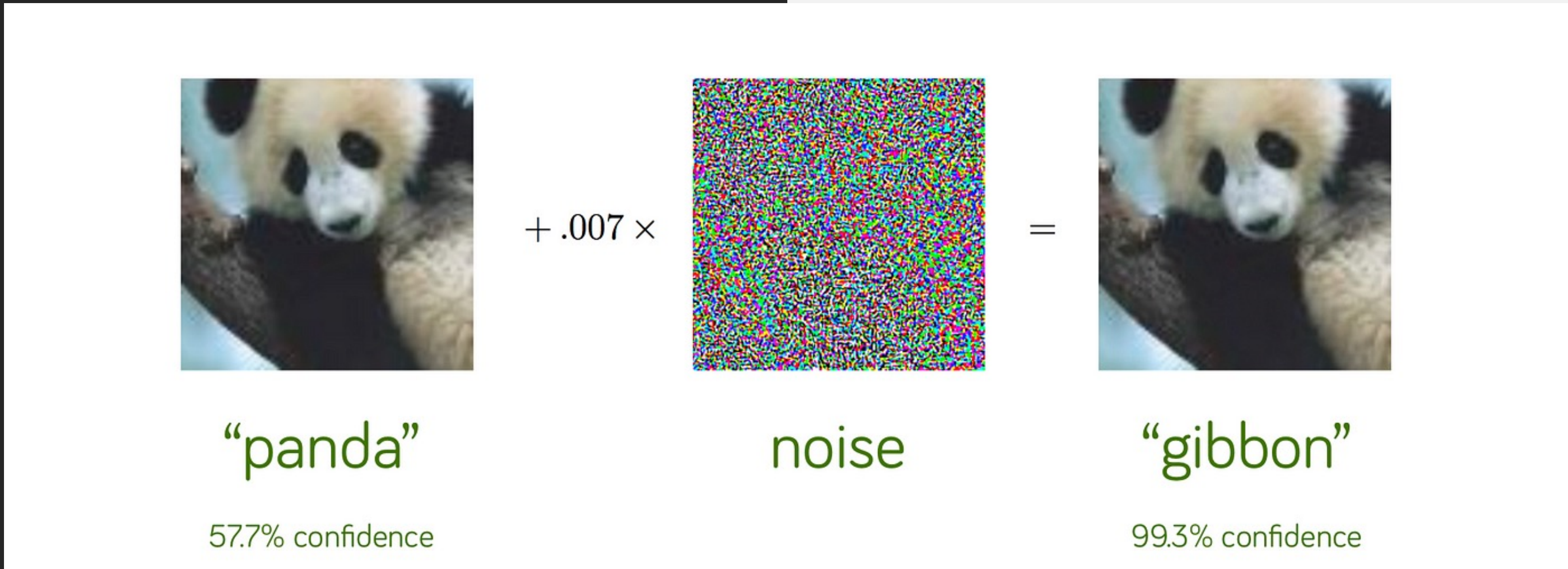
Would you trust a computer to make a judicial judgement if only trained on some types of cases?

Also:

Some images are hard to classify correctly

Machines can get it wrong

Identified as a 45mph speed sign



And can get confused by an 'adversarial attack'

Study this as a conditioning problem in numerical analysis

Machine learning models can memorise data: our privacy is at risk

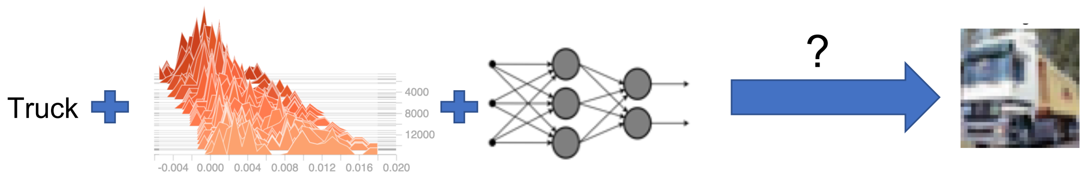
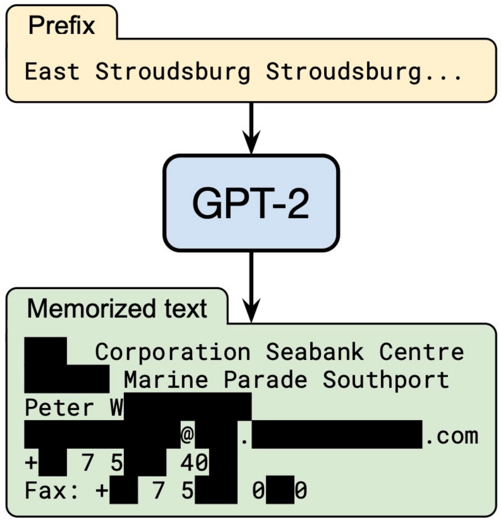


Image classification models can memorise their training images through the training updates (i.e. gradients).



Language models can also memorise their training
can be extracted using text prompts.

Chen C, Campbell N. Analysing Training-Data Leakage from Gradients through Linear Systems and Gradient Matching, The 33rd British Machine Vision Conference (BMVC), 2022

Carlini, Nicholas, et al. "Extracting training data from large language models." 30th USENIX Security Symposium (USENIX Security 21). 2021.

One way to mitigate the privacy risk: training with differential privacy